Lab 5 - Advanced Machine Learning

EM Algorithm

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General information

- **Assignment** : Alone or in pairs, you will code the algorithms you learnt in Ścikit-learn formalism’
- **Due** : Each assignment has to be sent at most 7 days after the lab session at pierre.houdouin@centralesupelec.fr
- **Grading** : There are 5 lab sessions, each lab session is worth 4 points, the average will count for half of your final grade
- **Questions** : If you have questions, comments or feedbacks about the lab session, feel free to contact me by email

Reminder on the EM algorithm

- Suppose we have $x_1, \ldots, x_n \in \mathbb{R}^2$ data points split in K different clusters coming from multivariate gaussian distributions.

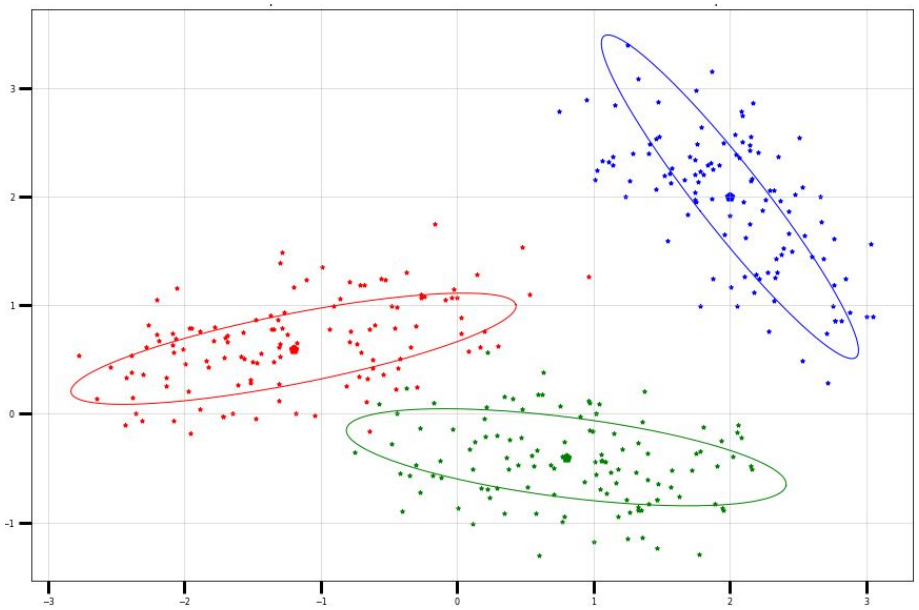


Figure 1: Points of $\mathbb{R}^2$ with $K = 3$

Reminder on the EM algorithm

- We don't know to which cluster each point belongs to, we only know that there are K possibilities.
- We want an algorithm that allows us to find the mean and covariance of each cluster and also guess the probability of membership for each data point.

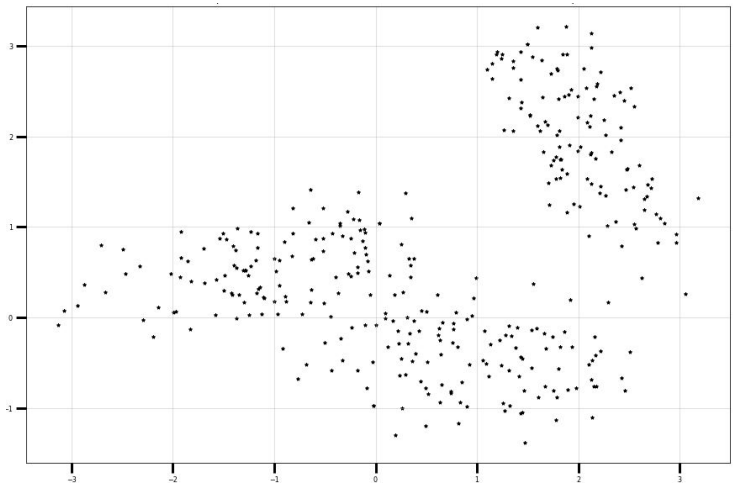


Figure 2: Clustering task

Reminder on the EM algorithm

Chicken and egg problem

- If we knew the label of each point, we could estimate $\hat{\mu}$ and $\hat{\Sigma}$ for each cluster.
- If we knew the parameters μ and Σ of each cluster, we could compute the probability of membership for each class using the Probability Density Function of the gaussian distribution.

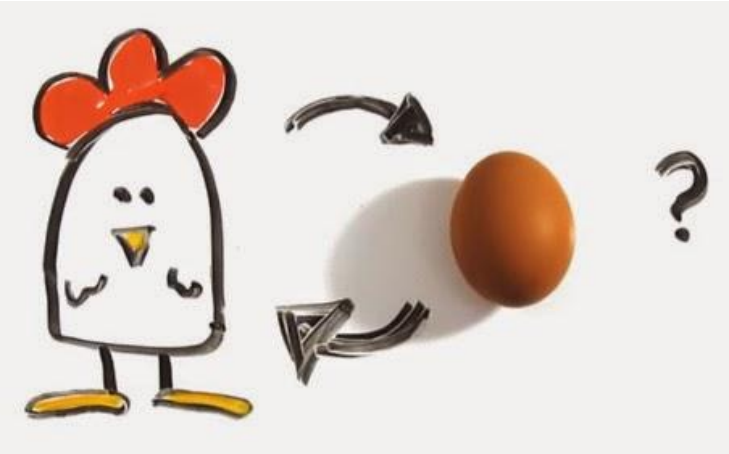


Figure 3: Chicken and egg problem

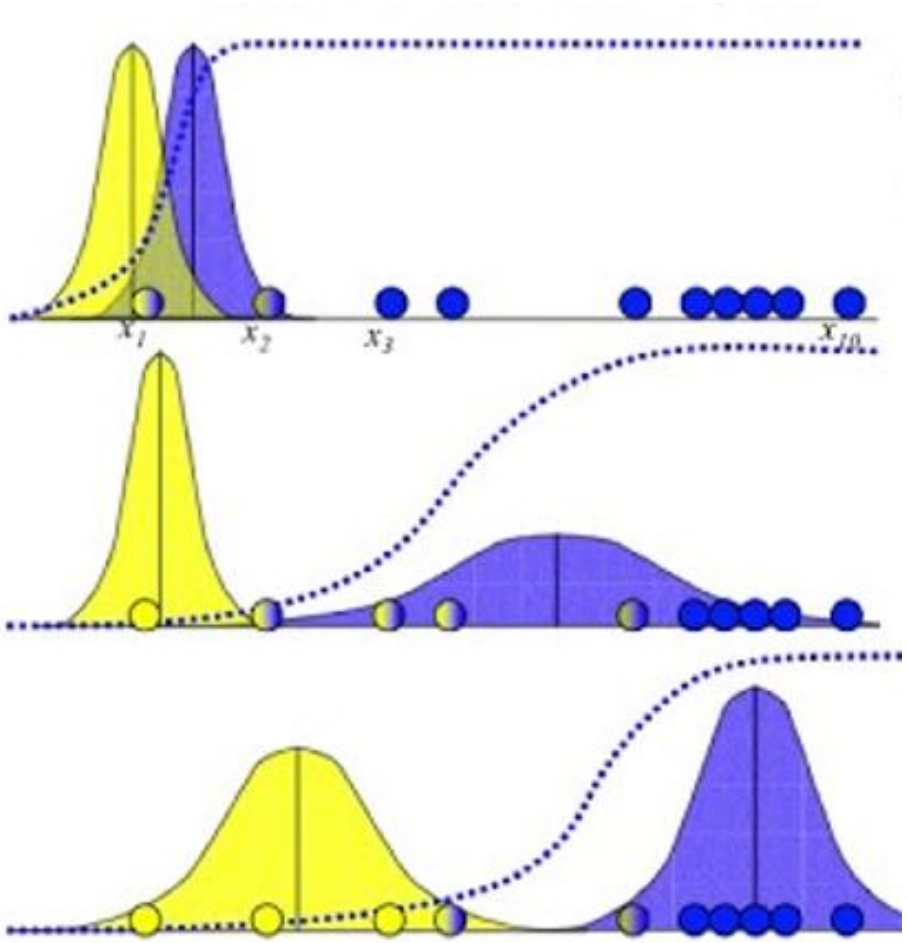
Reminder on the EM algorithm

Solution : Start by a random initialization and iterate, hoping it converges toward the solution

- We start with K randomly chosen gaussian distributions
- For each point, we compute the probability of membership for each class (Expectation part)
- Using the new membership probabilities that were just computed, we update the parameters of the gaussian distributions (Maximization of the likelihood part)

Theory on the EM algorithm ensures us that at each iteration, the expected value of the likelihood of the n-sample increases

Reminder on the EM algorithm



Back to the general case

Presentation of the problem

- We observe a n-sample $x_1, ..., x_n \in \mathbb{R}^p$, each observation following a multivariate gaussian distribution of probability density function g
- The data are extracted from K different distributions, each having prior π_k

Goal

We aim to find $\theta = (\pi_1, ..., \pi_K, \mu_1, ..., \mu_K, \Sigma_1, ..., \Sigma_K)$, the parameters of our gaussian mixture model so that we can obtain the probability density function for every point :

$$\forall x \in \mathbb{R}^p, f(x) = \sum_{i=k}^{K} \frac{\pi_k}{\sqrt{(2\pi)^p det(\Sigma_k)}} exp\left(-\frac{1}{2}(x-\mu_k)^T \Sigma_k^{-1}(x-\mu_k)\right)$$

Back to the general case

- We start by randomly initializing $\theta = \theta_0$
- Since the label of each observation is a latent unobserved variable, we can't compute the likelihood of the n-sample
- However, we can compute the membership probabilities, using θ_0, thus also the expected value of the likelihood

We look for θ_1 that maximizes the expected value of the log-likelihood : **M step**
Then we will compute again the membership probabilities using θ_1 : **E step**

$$\boxed{\theta_{n+1} = \arg\min_{\theta} \sum_{i=1}^{N} \sum_{k=1}^{K} \frac{\pi_k^n g(x_i, \mu_k^n, \Sigma_k^n)}{\sum_{j=1}^{K} \pi_j^n g(x_i, \mu_j^n, \Sigma_j^n)} \left(log(\pi_k) + log(g(x_i, \mu_k, \Sigma_k)\right)}$$

Question 1 : Optimization of the expected value of the log-likelihood

- 1) Find the expression of $\pi_1^{n+1}, \ldots, \pi_K^{n+1}$ using θ_n
- 2) Find the expression of $\mu_1^{n+1}, \ldots, \mu_K^{n+1}$ using θ_n
- 3) Find the expression of $\Sigma_1^{n+1}, \ldots, \Sigma_K^{n+1}$ using θ_n

Question 2 : Implementation of the EM algorithm

Implement your own EM algorithm and test it on simulated data (a data simulator is provided in the notebook). You may focus on the 2D case and plot the results to compare with the original labels.